

Approaches to HL-LHC Prompt vs Non-Prompt Muon Track Discrimination

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Motivation

- ▶ LHC ran at average pileup $\langle\mu\rangle \approx 50$ before shut-down
- ▶ HL-LHC: $\langle\mu\rangle \approx 200$ collisions per bunch crossing (2030s)
- ▶ Data from many more overlapping tracks per event
- ▶ Harder to separate muons originating from electroweak and Higgs boson decays (prompt) versus in the jets of hadron decays (non prompt)
- ▶ Motivates more sophisticated muon track discrimination tools
- ▶ Investigate BDT and Neural Network approaches

Background

- ▶ Prompt muons originate at primary vertex
 - ▶ Produced in electroweak and Higgs boson decays
- ▶ Non-prompt muons originate inside jets
 - ▶ Produced in b - and c -hadron decays
- ▶ Muon track isolation separates the two
 - ▶ Measured by the track activity surrounding the muon
 - ▶ Prompt muons appear isolated

Prompt vs Non-Prompt Leptons

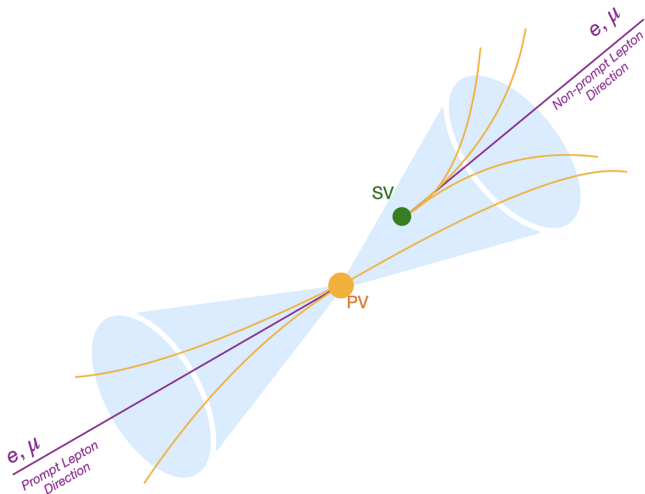
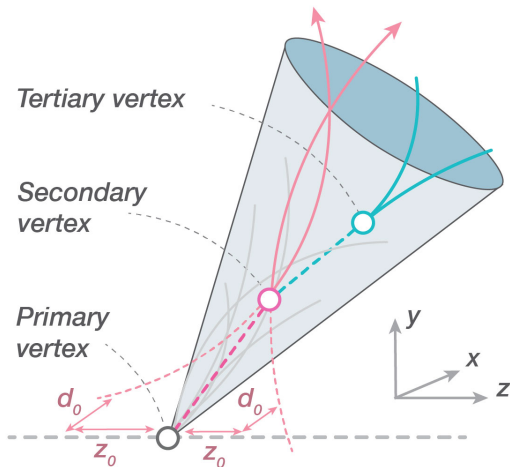


Illustration of prompt and non-prompt leptons, reproduced from ATLAS
Paper Draft MUON-2024-01

Jet Vertex Structure

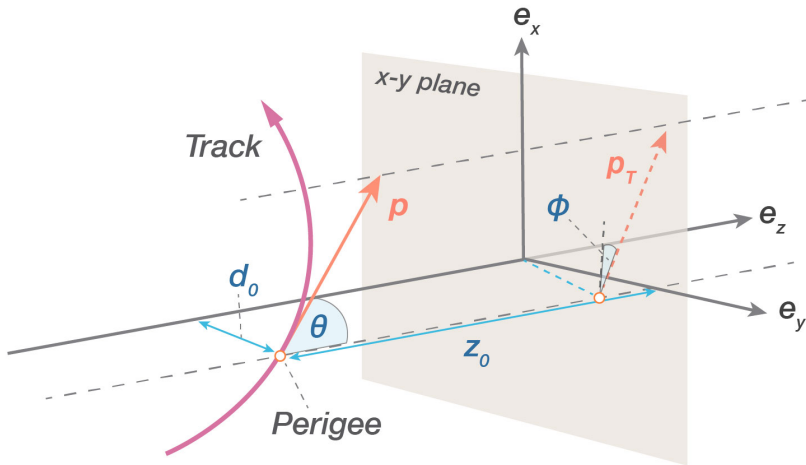


Displaced secondary and tertiary vertices inside a jet give non-prompt tracks large d_0 and z_0

Objectives

- ▶ Train machine learning classifiers to separate prompt from non-prompt muons based on simulated $\langle\mu\rangle = 200$ inner detector track information
- ▶ Compare performance against cut-based isolation baseline
- ▶ XGBoost-Based BDT Classifier Model
 - ▶ Muon kinematics with individual neighboring track features
- ▶ Neural Network Approach
 - ▶ PyTorch MLP with the same inputs and preprocessing as the BDT
- ▶ Cut-Based Isolation
 - ▶ Consider relative track isolation $\sum p_T^{\text{cone}}/p_T^\mu$ of muon by including nearby tracks in a ΔR and $|\Delta z_0 \sin \theta|$ based cone
 - ▶ Sweep isolation threshold to maximize signal/ background discrimination

Track Parametrization



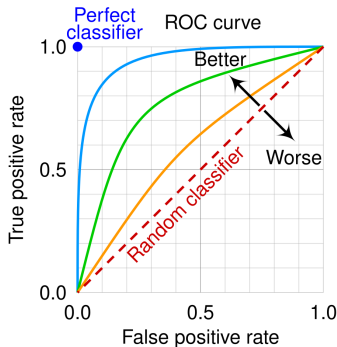
Track kinematics (momentum p , transverse momentum p_T , angles θ , ϕ) and impact parameters (transverse d_0 , longitudinal z_0) defined at the perigee

Methodology

- ▶ Signal uses a single prompt muon with pileup $\langle\mu\rangle = 200$
- ▶ Background uses non-prompt muons from b -hadron decays in $HH(4b)$ events
- ▶ Candidates are reconstructed muons, labeled by sample of origin
- ▶ Muon features are p_T , η , $|z_0 \sin \theta|$, d_0
- ▶ Neighboring track features
 - ▶ Up to $K = 10$ tracks within a ΔR and $|\Delta z_0 \sin \theta|$ cone
 - ▶ Each carrying d_0 , p_T , $\Delta\eta$, ΔR , $\Delta z_0 \sin \theta$
 - ▶ Slots filled in two orderings: ΔR -ascending and p_T -descending
- ▶ Tracks < 1 GeV are dropped
- ▶ Neighboring tracks limit to physical constraint $\Delta R < 0.5$
- ▶ Background reweighted to signal in muon (p_T, η) and corrected for class imbalance
- ▶ 80/20% split by event into training and holdout sets
 - ▶ Holdout reported for all quoted performance

Evaluation with ROC and AUC

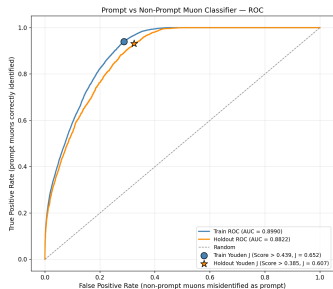
- ▶ ROC (Receiver-Operator Characteristic) curves
 - ▶ True Positive Rate vs False Positive Rate as the classification score threshold is swept
 - ▶ TPR = prompt muons correctly identified
 - ▶ FPR = non-prompt muons misidentified as prompt
 - ▶ A better classifier bows toward the top-left corner
- ▶ AUC = area under curve (0.5 = random, 1.0 = perfect)
- ▶ Youden's $J = \text{TPR} - \text{FPR}$



Results

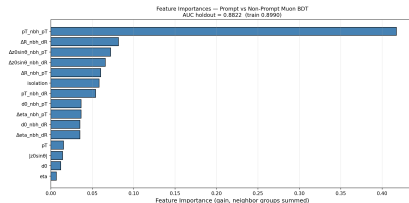
- ▶ BDT reaches a holdout AUC of 0.8822 at the physical cone
- ▶ Neural network reaches 0.8604 on the same inputs and split
- ▶ Both outperform the best cut-based isolation at 0.7740
- ▶ Prompt vs Non-Prompt discrimination carried primarily by the data of surrounding track activity
- ▶ Preliminary results at relatively small sample size show direction is promising
- ▶ Cut-based isolation improves with $|\Delta z_0 \sin \theta|$ cut chosen per track in isolation calculation based on $|\eta|$ and p_T values
- ▶ Findings presented at CERN's Isolation and Fakes Forum in July 2026

Best BDT Result



```

RAW SETTINGS (muonBDT.py v1.8)
Samples      : signal+non_prompt+non_prompt, bkg+non_bkg (non-prompt): label=provenance
Candidates  : muons (16000): pt = 1.0 GeV; bkg all muons/event; signal 1/event
Features     : mean pt, eta, [20:100] / # (10:100), isolation, neighbors carry db, pt, eta, DR, 800:100
Isolation    : 3pt(core tracks)/pt(muons), cone = db+3 and [20:100] track+delta muon+10 db mu, pt=1 def, muon excluded
Neighbors    : ~40:40 same-event tracks, db+3 and [20:100] track+delta muon (signed vertex sep); same cone train/holdout; db+acc & pt+acc; NNN padded
Reweighting  : G2Muweighter (pt,eta) bkg=signal, train only
Imbalance    : scale_gm_weighted: 50 on signal
Events       : signal 9,291 events / 9,291 muons, background 9,999 events / 370,684 muons
Trains/holds : 465,378 / 114,897 muons (50:50 split by event)
Hyperparams  : trees=200, depth=3, lr=0.1, subsample=0.8, colsample=0.8, min_child_weight=1.5, reg_lambda=1.5
    
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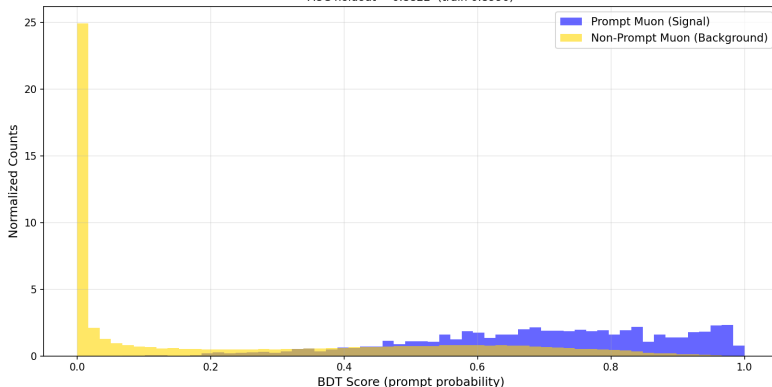
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neighbor feature key table - up to 60:60 tracks (in the neighbor cone, same event)
~mean_db+acc - ranked by db ascending (lowest first)
~cone_db+pt - ranked by pt ascending (lowest first)
each db+acc = sum of gsis over all N slots of that (var, weighting)

RAW SETTINGS (muonBDT.py v1.8)
Samples      : signal+non_prompt+non_prompt, bkg+non_bkg (non-prompt): label=provenance
Candidates  : muons (16000): pt = 1.0 GeV; bkg all muons/event; signal 1/event
Features     : mean pt, eta, [20:100] / # (10:100), isolation, neighbors carry db, pt, eta, DR, 800:100
Isolation    : 3pt(core tracks)/pt(muons), cone = db+3 and [20:100] track+delta muon+10 db mu, pt=1 def, muon excluded
Neighbors    : ~40:40 same-event tracks, db+3 and [20:100] track+delta muon (signed vertex sep); same cone train/holdout; db+acc & pt+acc; NNN padded
Reweighting  : G2Muweighter (pt,eta) bkg=signal, train only
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```

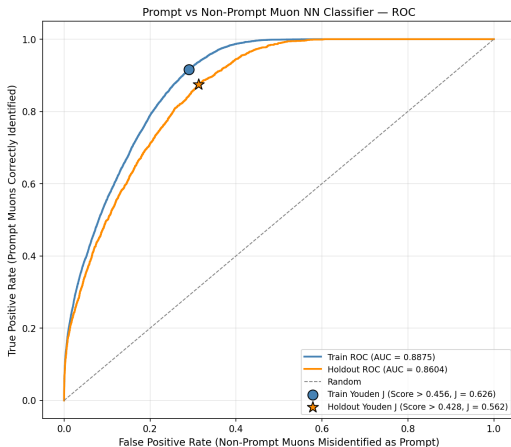
BDT Output Score

Holdout BDT Score — Prompt vs Non-Prompt Muons
AUC holdout = 0.8822 (train 0.8990)



```
RUN SETTINGS      (muonBDT.py v1.0)
Samples          : signal=run_invPtPU200 (prompt); bkg=run_bjet (non-prompt); label=provenance
Candidates       : muons (isMuon), pT >= 1.0 GeV; bkg all muons/event, signal l/event
Features         : muon pT, eta, |z0sin0|, d0 (signed), isolation; neighbors carry d0, pt, Δeta, ΔR, Δz0sin0
Isolation        : ΔpT(cone tracks)/pT(muon); cone = dR<0.5 and |z0sin0_track-z0sin0_muon|<10.0mm, pT>=1 GeV, muon excluded
Neighbors        : <=K=10 same-event tracks, dR<0.5 and |Δz0sin0|<10.0mm (signed vertex sep); same cone train/holdout; dR-asc & pT-desc; NaN-padded
Reweighting      : GBReweighter (pT,eta) bkg->signal, train only
Imbalance        : scale_pos_weight=01.62 on signal
Events           : signal 9,291 events / 9,291 muons, background 9,999 events / 570,694 muons
Train/holdout    : 465,378 / 114,607 muons (80/20 split by event)
Hyperparams      : trees=200, depth=3, lr=0.1, subsample=0.8, colsample=0.8, min_child_weight=1.0, reg_lambda=1.0
```

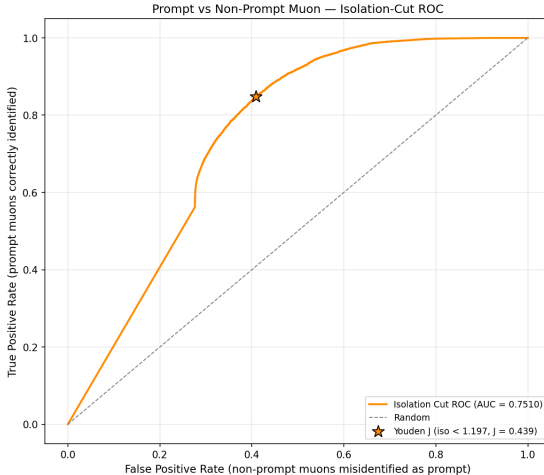
Best Neural Network Result



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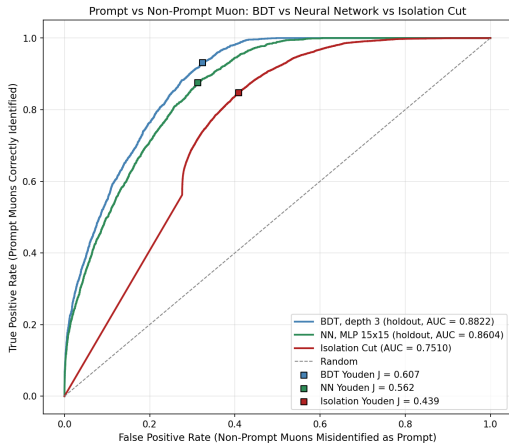
RUN SETTINGS (muonNN.py v1.0)
Samples      : signal=run_InvPtPU280 (prompt), bkg=run_bjet (non-prompt); label=provenance
Candidates   : muons (isMuon), pT >= 1.0 GeV; bkg all muons/event. signal 1/event
Features      : muon pT, eta, |z0sinθ|, d0 (signed), isolation; neighbors carry d0, pT, Δeta, ΔR, Δz0sinθ [d0-mode=both]
Neighbors     : <=K=10 same-event tracks, dR<0.5 and |Δz0sinθ|<10mm; dR-asc, pT-desc; zero-padded
Network       : MLP 105->15->15->1, ReLU, 1846 parameters
Training      : Adam lr=0.001, batch=512, epochs=21; inputs standardized with training statistics
Reweightings  : GBReweighter (pT,eta) bkg->signal, train only
Train/holdout : 465,378 / 114,607 muons (80/20 split by event)
    
```

Best Cut-Based Isolation Result



```
RUN SETTINGS (isoPLOT.py v1.0)
Samples      : signal=run_InvPtPU200 (prompt), bkg=run_bjet (non-prompt); label=provenance
Variable     : relative track isolation = (sum pt of cone tracks) / pt(muon)
Cone         : DR < 0.4 and |z0sinθ_track - z0sinθ_muon| < auto(pt,|eta)| table; tracks pT ≥ 1.0 GeV, muon excluded
Muons        : signal 9,291 (9,291 events), background 570,694 (9,999 events)
```

Three Approaches Comparison

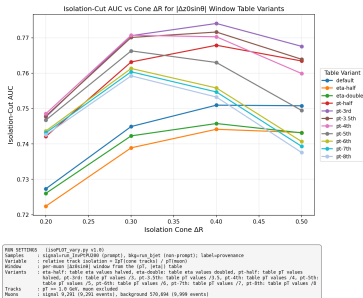
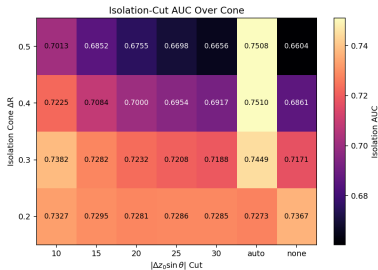


```

RUN SETTINGS
Samples : signal=run_InvPtPU200 (prompt), bkg=run_bjet (non-prompt); label=provenance
Cone (ML) : dR<0.5 and |Δz0sinθ|<10mm, K=10 neighbors, d0-mode=both, isolation on
BDT : XGBoost trees=200, depth=3, lr=0.1; NaN-padded neighbor slots
NN : MLP 105->15->15->1, ReLU, Adam lr=0.001, 21 epochs; zero-padded, standardized
Isolation : IPt(cone)/PT(muon), best cone dR<0.4 with 'auto' |Δz0sinθ| window
            'auto' = per-track window from the (track pT, |eta|) table
Reweighting : GBReweighter (pT,eta) bkg->signal, train only; scale_pos_weight on signal
Evaluation : BDT and NN on the same 20% holdout (split by event); Isolation is untrained
Youden J : box-marked points at max(TPR-FPR) on each curve
    
```

NIU ATLAS Group Window Table

- Standard isolation sweeps one fixed $|\Delta z_0 \sin \theta|$ cut applied to every track in the cone
- NIU ATLAS group's table instead assigns each track its own window from that track's p_T and $|\eta|$
- Optimized table by testing variants with all p_T (or $|\eta|$) bin boundaries divided or multiplied by a constant
- Best variant divides the p_T boundaries by 3, giving each track a wider window, raising AUC from 0.7510 to 0.7740



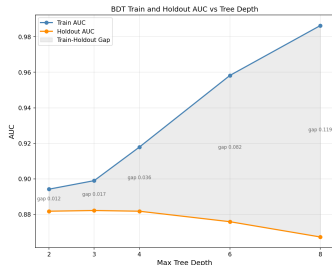
Best Performing Window Table

Per-track $|\Delta z_0 \sin \theta|$ window [mm], p_T bin values divided by 3

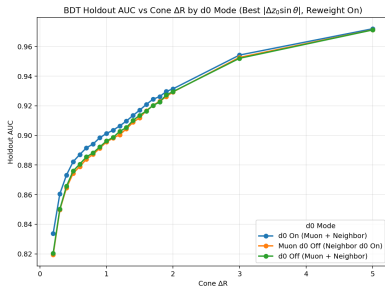
Track p_T range [GeV]	$ \eta < 1$	$1 < \eta < 2$	$ \eta > 2$
1.00–1.33	2.0	1.6	1.0
1.33–1.67	3.3	2.5	1.6
1.67–2.00	4.2	3.3	2.5
2.00–2.33	5.4	5.2	4.2
2.33–3.33	6.0	5.2	4.2
3.33–6.67	6.0	5.5	4.2
6.67–10.00	6.0	6.6	5.4
> 10.00	6.0	8.5	8.5

Challenges

- ▶ Overfitting from training data memorization
- ▶ Standardizing multifarious information
- ▶ Running hundreds of configurations
- ▶ Nonphysical cone artefacts
- ▶ Neural network optimization



Run SETTINGS
Samples : signal:run_inspP200 (prompt), bkg:run_jet (non-prompt); label:performance
Core : d0:1 and 1d0:100-100; K10: neighbors; d0: neighbors; isolation on
Reweighting : GDBNewweight (PT,eta) bkg:signal, train only; scale_pos_weight on signal
Selection : best holdout AUC per tree depth across d20 d00n scale (Full statistics)



Future Directions

- ▶ Train on physically simulated samples
 - ▶ Current signal adds the prompt muon directly without a simulated decay event
 - ▶ Prompt muons emerging from actual W , Z or Higgs decays instead
- ▶ Greater sample size
 - ▶ Only $\sim 7,500$ signal muons currently available for training
 - ▶ Would permit deeper trees and larger networks before overfitting sets in
- ▶ Increase the number of neighboring tracks beyond $K = 10$
- ▶ Incorporate calorimeter data alongside inner detector tracks
 - ▶ Transverse energy in a cone around the muon; standard in ATLAS isolation tools
- ▶ Develop the BDT and Neural Network to experimentally-useful discrimination levels
 - ▶ Evaluated as non-prompt rejection at fixed prompt efficiency working points
 - ▶ For eventual comparison against ATLAS isolation taggers

Acknowledgements

- ▶ US-ATLAS SUPER
- ▶ Jahred Adelman
- ▶ Kevin Sedlaczek
- ▶ NIU ATLAS Group
- ▶ NIU METIS HPC
- ▶ Slide 3 illustration reproduced from ATLAS Paper Draft MUON-2024-01 (unpublished)
- ▶ Code and results for this project are on GitHub:
 - ▶ <https://github.com/sergeiuzunian/muon-isolation>

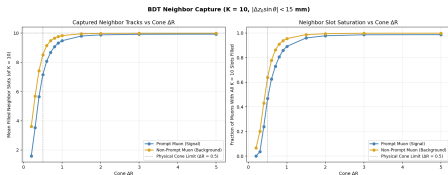
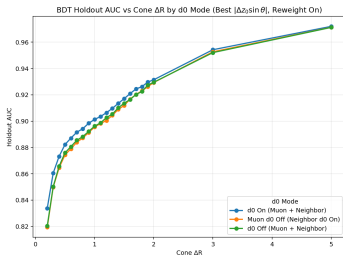
Backup Slides

Managing the Optimization Process

- ▶ Keeping track of collected data throughout development
 - ▶ Each run is one train-and-evaluate cycle at one choice of settings
 - ▶ Many interacting settings: cone cuts, d_0 modes, reweighting, neural network architecture
 - ▶ Output of two runs indistinguishable without a record of its settings
- ▶ Addressed by the addition of a standardized run-settings box added to every plot
 - ▶ Every run also writes a summary file and a settings-encoded output directory
- ▶ Requiring hundreds of runs to scan the parameter space
 - ▶ Over 500 BDT runs across cone, d_0 , and hyperparameter scans
 - ▶ Full statistics runs can take hours each
- ▶ Addressed with batch processing on the NIU METIS cluster
 - ▶ Employed reusable and tunable PBS scripts
 - ▶ Scripts build features once and reuse them across configurations

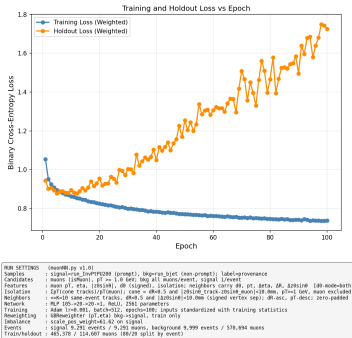
Nonphysical Cone Artifact

- ▶ Holdout AUC rising with cone ΔR and never plateauing
 - ▶ Reaches ≈ 0.97 once the cone spans the whole event
 - ▶ Cannot reflect genuine isolation physics
 - ▶ Current hypothesis: widening the cone fills more of the K neighbor slots
- ▶ Mean filled slots rise from ~ 2 at $\Delta R = 0.2$ to all $K = 10$ by $\Delta R \approx 1.5$
 - ▶ A wider cone supplies more neighboring track information
- ▶ Addressed by restricting the analysis to the $\Delta R \leq 0.5$ “physical” approximation

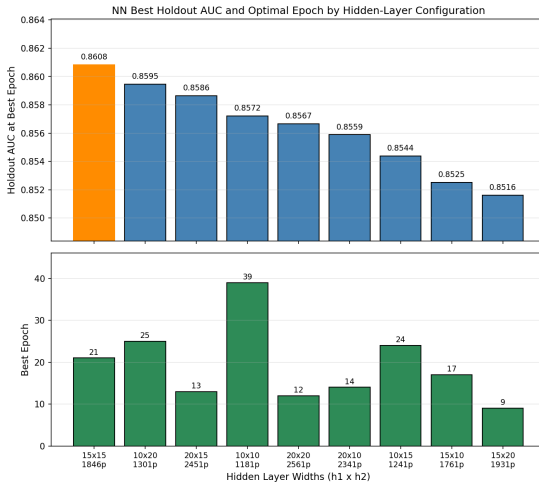


Neural Network Training

- ▶ Overfitting within a few epochs
 - ▶ Only $\sim 7,400$ signal muons available for training
 - ▶ Holdout loss turns upward while training loss keeps falling
 - ▶ Addressed by tracking holdout loss and AUC at every epoch
 - ▶ Epoch of maximum holdout AUC taken as the stopping point
- ▶ Employed a small network of $\sim 1,850$ parameters



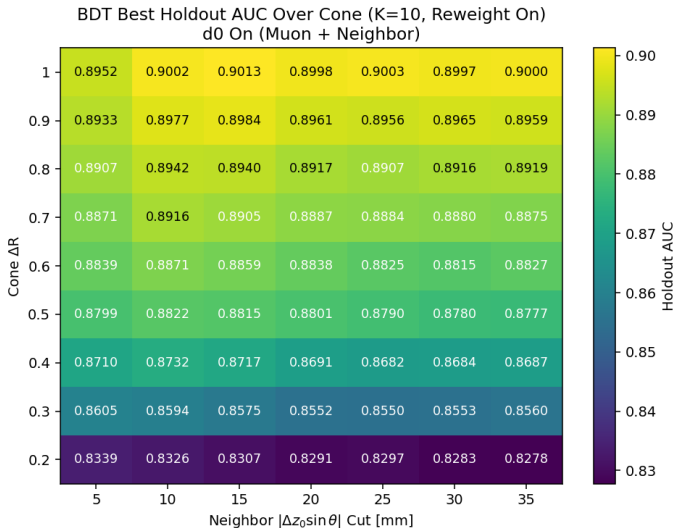
Neural Network Architecture Scan



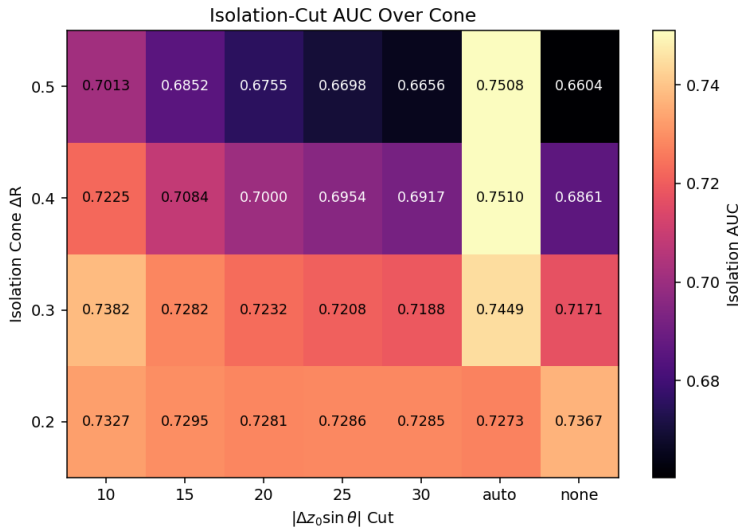
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RUN SETTINGS (muonNN.py)
Samples : signal=run_inyPtPU200 (prompt), bkg=run_bjet (non-prompt); label=provenance
Cone : GR=8.5 and |Δz|sinθ|<10.6mm, K=10 neighBors, d0-node=both, isolation on
Training : Adam lr=0.001, batch=512, up to 40 epochs; inputs standardized (train stats)
Reweightig : GBReweighter (pt,eta) bkg->signal, train only; scale_pos_weight on signal
Best epoch : epoch of maximum holdout AUC (selected on holdout)
    
```

BDT Cone Scan



Isolation Cone Scan



Isolation Window Optimization

- ▶ Cut-based isolation uses a per-track $|\Delta z_0 \sin \theta|$ window binned in p_T and $|\eta|$
- ▶ Optimized the NIU ATLAS group's original table by testing variants with all p_T (or $|\eta|$) bin boundaries divided or multiplied by a constant
- ▶ Best variant divides the p_T boundaries by 3, assigning each track a wider window
- ▶ Increased AUC from 0.7510 to 0.7740

